

Consider the vectors and a matrix below:

$$\begin{aligned}x^T &= (-10, -3, 3)^T \\ y^T &= (9, 0, -7)^T \\ A &= \begin{pmatrix} 1 & 4 & 2 \\ 4 & -3 & 4 \end{pmatrix}\end{aligned}$$

The value of $x^T y$: -111

If $z = 3x - 5y$, then $z_1 = -78$

If $z = Ax$, then $z_2 = -19$

If $C = AA^T$, then $C_{1,1} = 21$

$$x = \begin{pmatrix} -10 \\ -3 \\ 3 \end{pmatrix} \quad y = \begin{pmatrix} 9 \\ 0 \\ -7 \end{pmatrix}$$

$$\textcircled{1} x^T y = (-10 \quad -3 \quad 3) \begin{pmatrix} 9 \\ 0 \\ -7 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \end{pmatrix} = -90 + 0 - 21 = -111$$

$$\textcircled{2} 3x - 5y = 3 \begin{pmatrix} -10 \\ -3 \\ 3 \end{pmatrix} - 5 \begin{pmatrix} 9 \\ 0 \\ -7 \end{pmatrix} = \begin{pmatrix} z_1 \\ \cdot \\ \cdot \end{pmatrix} = -30 - 45 = -78$$

$$\textcircled{3} Ax = \begin{pmatrix} 1 & 4 & 2 \\ 4 & -3 & 4 \end{pmatrix} \begin{pmatrix} -10 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} \cdot \\ z_2 \end{pmatrix} = -40 + 9 + 6 = -19$$

$$\textcircled{4} AA^T = \begin{pmatrix} 1 & 4 & 2 \\ 4 & -3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 4 & -3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} c_{11} & \cdot \\ \cdot & \cdot \end{pmatrix} = 1 + 16 + 4 = 21$$

Give the value of α , if we know that x and y are linearly dependent.

$$x = \begin{pmatrix} 6 \\ 9 \end{pmatrix}, \quad y = \begin{pmatrix} 2 \\ \alpha \end{pmatrix}$$

Answer: 3

If

$$A = \begin{pmatrix} -3 & 2 & -2 \\ 1 & -4 & -1 \\ 4 & 0 & -3 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & -5 \\ -4 & 4 \end{pmatrix},$$

then $\det(A) = -70$ and $\det(B) = -4$

$$\det(B) = 16 - 20 = -4$$

$$\det(A) = \begin{vmatrix} -3 & 2 & -2 & -3 & 2 \\ 1 & -4 & -1 & 1 & -4 \\ 4 & 0 & -3 & 4 & 0 \end{vmatrix} = -36 - 8 + 0 - 32 - 0 + 6 = -70$$

Using some matrices A and B , and some column vectors u and v we obtained, that

$$AB + uv^T = \begin{pmatrix} 2 & -4 & 1 & -4 & -5 & -5 \\ -3 & 1 & 2 & -3 & -3 & -5 \\ -1 & 3 & 3 & -4 & 5 & -4 \end{pmatrix}.$$

Mark the true statements.

Select one or more:

☒ a. Matrices A and B have the same number of rows.

☒ b. B has as many columns as many coordinates v has.

☒ c. The product Bv is computable.

☒ d. The product Au is computable.

☒ e. The matrix B has 6 columns.

Mark the computable expressions.

$$A = \begin{pmatrix} 4 & -5 & 4 & -4 & -1 & -2 \\ -3 & -1 & 3 & 2 & -3 & 1 \\ -2 & 5 & 2 & 4 & -2 & 2 \\ -1 & -5 & 5 & -1 & 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 & 3 & -4 & -4 & 2 \\ 2 & -4 & -4 & 2 & 3 & 3 \\ -2 & -4 & 3 & 2 & 1 & 4 \\ -1 & -2 & 4 & -5 & 4 & -1 \\ -2 & 2 & -5 & 3 & 3 & 4 \\ 4 & 4 & -1 & 3 & -1 & 2 \end{pmatrix}, \quad u = \begin{pmatrix} -4 \\ 2 \\ -5 \\ -1 \\ 1 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} -5 \\ -4 \\ -1 \\ 1 \\ 1 \end{pmatrix}.$$

Select one or more:

☒ a. Bu 6×6 6×1

☒ b. uv^T 6×1 1×4

☒ c. $v^T A$ 1×4 4×6

☒ d. $u^T v^T$ 1×6 1×4

☒ e. uB 6×1 6×6

Mark the sets of vectors which span $\mathbb{R}^2$.

$\mathbb{R}^2 \rightarrow$ 2次元空間

☒ a. $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

☒ b. $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

☒ c. $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

☒ d. $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

a) $\lambda_1 v_1 + \lambda_2 v_2 = 0$
 $\begin{cases} \lambda_1 = 0 \\ \lambda_1 + \lambda_2 = 0 \end{cases} \rightarrow \lambda_2 = 0$ v_1, v_2 are linearly independent. $\rightarrow \mathbb{R}^2$

b) 1 linearly independent $\rightarrow \mathbb{R}^1$

c) dependent

d) $v_1 = v_2 + v_3$ v_2, v_3 are linearly independent $\rightarrow \mathbb{R}^2$

Mark the sets of vectors which are linearly independent in $\mathbb{R}^3$.

linearly independent $\rightarrow$ 3次元空間
 $\mathbb{R}^1$ $\mathbb{R}^2$ $\mathbb{R}^3$ は関係ない。

☒ a. $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ -2 \\ 0 \end{pmatrix}$

☒ b. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

☒ c. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

☒ d. $v_1 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

a) $v_2 = -2v_1$ dependent

b) 2 linearly independent $\rightarrow \mathbb{R}^2$

c) 3 linearly independent $\rightarrow \mathbb{R}^3$

d) 1 linearly independent $\rightarrow \mathbb{R}^1$

Mark the vectors which can be obtained as a linear combination of

Time left 0:58:21

$$v_1 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}$$

☒ a. $\begin{pmatrix} -7 \\ 1 \\ 0 \end{pmatrix}$

☒ b. $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$

☒ c. $\begin{pmatrix} 0 \\ 9 \\ 0 \end{pmatrix}$

☒ d. $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

☒ e. $\begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix}$

$$\lambda_1 \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \end{pmatrix}$$

a) $\begin{cases} 2\lambda_1 + \lambda_2 + 3\lambda_3 = -7 \\ 2\lambda_2 - \lambda_3 = 1 \\ 0 = 0 \end{cases} \quad \begin{aligned} \lambda_3 &= 2\lambda_2 \\ 2\lambda_1 + \lambda_2 + 6\lambda_2 &= -7\lambda_2 + 1 \end{aligned}$

Mark the sets of vectors which are linearly independent in $\mathbb{R}^3$.

- ☒ a. $v_1 = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$
- ☒ b. $v_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$
- ☒ c. $v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix},$
- ☒ d. $v_1 = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix},$

Mark the computable expressions.

$A = \begin{pmatrix} 1 & 4 & 3 & -3 \\ 1 & -5 & 1 & -4 \\ -3 & -4 & -5 & 5 \\ 2 & 2 & 2 & 2 \end{pmatrix},$
4.4

$B = \begin{pmatrix} 5 & 3 & -5 & -1 \\ 2 & 3 & -3 & 3 \\ 1 & -4 & -4 & -2 \\ 3 & -4 & 3 & -1 \\ -2 & -5 & -4 & 4 \end{pmatrix},$
5.4

$u = \begin{pmatrix} 4 \\ 4 \\ 3 \\ -4 \end{pmatrix},$
4.1

$v = \begin{pmatrix} 4 \\ -3 \\ -3 \\ 5 \\ -2 \end{pmatrix}.$
5.1

Select one or more:

- ☒ a. AB^T 4.4 x 4.5
- ☒ b. v^TB 1.5 x 5.4
- ☒ c. uv^T 4.1 x 1.5
- ☒ d. uB 4.1 x 5.4
- ☒ e. uv

Mark the vectors which can be obtained as a linear combination of

- $v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, v_2 = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}, v_3 = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}$
- ☒ a. $\begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}$
- ☒ b. $\begin{pmatrix} -3 \\ -6 \\ -9 \end{pmatrix}$ $\begin{pmatrix} -3 \\ -6 \\ -9 \end{pmatrix} \times 2$ $\begin{pmatrix} -3 \\ -6 \\ -9 \end{pmatrix} \times 3$
- ☒ c. $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$
- ☒ d. $\begin{pmatrix} -7 \\ 0 \\ 0 \end{pmatrix} \leftarrow \times$ $\begin{pmatrix} -7 \\ 0 \\ 0 \end{pmatrix} \leftarrow \times$
- ☒ e. $\begin{pmatrix} 5 \\ 10 \\ 15 \end{pmatrix} \begin{pmatrix} 5 \\ 10 \\ 15 \end{pmatrix} \times 2$ $\begin{pmatrix} 5 \\ 10 \\ 15 \end{pmatrix} \times 3$

$$\lambda_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + \lambda_3 \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} = \begin{pmatrix} x \\ 2x \\ 3x \end{pmatrix}$$

Mark the sets of vectors which span $\mathbb{R}^2$.

- ☒ a. $v_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$
- ☒ b. $v_1 = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, v_2 = \begin{pmatrix} 3 \\ -2 \end{pmatrix},$
- ☒ c. $v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, v_2 = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$
- ☒ d. $v_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, v_2 = \begin{pmatrix} -4 \\ 2 \end{pmatrix},$

Using some matrices A and B , and some column vectors u and v we obtained, that

$$AB + uv^T = \begin{pmatrix} -3 & -4 & -3 & 5 & 2 & -3 \\ -5 & -1 & 4 & 3 & 5 & 4 \\ 5 & -5 & -4 & 5 & 4 & 1 \\ 3 & 1 & -4 & -5 & -2 & 3 \end{pmatrix}.$$

Mark the true statements.

4.6

Select one or more:

- ☒ a. The matrix A has as many columns as many rows B has.
- ☒ b. The vector v has 6 coordinates.
- ☒ c. The product Bv is computable.
- ☒ d. The vector u has 6 coordinates.
- ☒ e. The matrix B has 4 rows.

AB
4.6
 $\swarrow$
 $A \times B$
4.0 0.6

uv^T
4.6
 $\swarrow$
 $u \times v^T$
4.0 0.6
 $\downarrow$
 v
6.0

If

$$A = \begin{pmatrix} -4 & -3 & 4 \\ 2 & -2 & -2 \\ -3 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -3 & 5 \\ -5 & 4 \end{pmatrix},$$

then $\det(A) = -28$ and $\det(B) = 13$

$$\det(B) = (-12) - (-25) = -12 + 25 = 13$$

$$\det(A) = \begin{vmatrix} -4 & -3 & 4 & -4 & 3 \\ 2 & -2 & -2 & 2 & -2 \\ -3 & 0 & 1 & -3 & 0 \end{vmatrix} = 8 - 18 + 0 - 24 - 0 + 6 = -28$$

Give the value of α , if we know that x and y are linearly dependent.

$$x = \begin{pmatrix} -6 \\ 12 \end{pmatrix}, y = \begin{pmatrix} 3 \\ \alpha \end{pmatrix}$$

$$12 \div (-2) = -6$$

Answer: -6

Consider the vectors and a matrix below:

$$\begin{aligned}x^T &= (-6, 8, 6, 9)^T \\ y^T &= (-2, 9, -3, -7)^T \\ A &= \begin{pmatrix} 3 & 1 & 0 & 0 \\ -1 & -2 & -1 & -4 \end{pmatrix}\end{aligned}$$

The value of $x^T y$:

If $z = 6x - 2y$, then $z_1 =$

If $z = Ax$, then $z_2 =$

If $C = AA^T$, then $C_{2,2} =$

$$x = \begin{pmatrix} -6 \\ 8 \\ 6 \\ 9 \end{pmatrix} \quad y = \begin{pmatrix} -2 \\ 9 \\ -3 \\ -7 \end{pmatrix}$$

$$x^T y = (-6 \ 8 \ 6 \ 9) \begin{pmatrix} -2 \\ 9 \\ -3 \\ -7 \end{pmatrix} = (\cdot) = 12 + 72 - 18 - 63 = 3$$

$$6x - 2y = \begin{pmatrix} -36 \\ 48 \\ 36 \\ 54 \end{pmatrix} + \begin{pmatrix} 4 \\ -18 \\ 6 \\ 14 \end{pmatrix} = \begin{pmatrix} z_1 \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = -36 + 4 = -32$$

$$Ax = \begin{pmatrix} 3 & 1 & 0 & 0 \\ -1 & -2 & -1 & -4 \end{pmatrix} \begin{pmatrix} -6 \\ 8 \\ 6 \\ 9 \end{pmatrix} = \begin{pmatrix} \cdot \\ z_2 \end{pmatrix} = 6 - 16 - 6 - 36 = -52$$

$$AA^T = \begin{pmatrix} 3 & 1 & 0 & 0 \\ -1 & -2 & -1 & -4 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & -2 \\ 0 & -1 \\ 0 & -4 \end{pmatrix} = \begin{pmatrix} \cdot & \cdot \\ \cdot & c_{22} \end{pmatrix} = 1 + 4 + 1 + 16 = 22$$

Consider the vectors and a matrix below:

$$\begin{aligned}x^T &= (4, 5, 4, 2)^T \\ y^T &= (-3, 0, -1, -9)^T \\ A &= \begin{pmatrix} -3 & 1 & -5 & -5 \\ 4 & 1 & 5 & 0 \end{pmatrix}\end{aligned}$$

The value of $x^T y$: -34

If $z = 4x - 5y$, then $z_3 =$ 21

If $z = Ax$, then $z_1 =$ -37

If $C = AA^T$, then $C_{1,1} =$ 60

$$x^T y = (4 \ 5 \ 4 \ 2) \begin{pmatrix} -3 \\ 0 \\ -1 \\ -9 \end{pmatrix} = (\cdot) = -12 + 0 - 4 - 18 = -34$$

$$4x - 5y = \begin{pmatrix} 16 \\ 20 \\ 16 \\ 8 \end{pmatrix} + \begin{pmatrix} 15 \\ 0 \\ 5 \\ 45 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = \begin{pmatrix} 16+5 \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = \begin{pmatrix} 21 \\ \cdot \\ \cdot \\ \cdot \end{pmatrix}$$

$$Ax = \begin{pmatrix} -3 & 1 & -5 & -5 \\ 4 & 1 & 5 & 0 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} z_1 \\ \cdot \end{pmatrix} = \begin{pmatrix} -12+5-20-10 \\ \cdot \end{pmatrix} = \begin{pmatrix} -37 \\ \cdot \end{pmatrix}$$

$$AA^T = \begin{pmatrix} -3 & 1 & -5 & -5 \\ 4 & 1 & 5 & 0 \end{pmatrix} \begin{pmatrix} -3 \\ 1 \\ -5 \\ -5 \end{pmatrix} = \begin{pmatrix} C_{11} & \cdot \\ \cdot & \cdot \end{pmatrix} = \begin{pmatrix} 9+1+25+25 \\ \cdot & \cdot \end{pmatrix} = \begin{pmatrix} 60 \\ \cdot & \cdot \end{pmatrix}$$

Mark the computable expressions.

$$A = \begin{pmatrix} 4 & -5 & 4 & -4 & -1 & -2 \\ -3 & -1 & 3 & 2 & -3 & 1 \\ -2 & 5 & 2 & 4 & -2 & 2 \\ -1 & -5 & 5 & -1 & 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 & 3 & -4 & -4 & 2 \\ 2 & -4 & -4 & 2 & 3 & 3 \\ -2 & -4 & 3 & 2 & 1 & 4 \\ -1 & -2 & 4 & -5 & 4 & -1 \\ -2 & 2 & -5 & 3 & 3 & 4 \\ 4 & 4 & -1 & 3 & -1 & 2 \end{pmatrix}, \quad u = \begin{pmatrix} -4 \\ 2 \\ -5 \\ -1 \\ 1 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} -5 \\ -4 \\ -1 \\ 1 \end{pmatrix}$$

Select one or more:

- ☒ a. uv^T $6 \cdot 1 \times 1 \cdot 4$
- ☒ b. $u^T v^T$ $1 \cdot 6 \times 1 \cdot 4$
- ☒ c. uB $6 \cdot 1 \times 6 \cdot 6$
- ☒ d. Bu $6 \cdot 6 \times 6 \cdot 1$
- ☒ e. $v^T A$ $1 \cdot 4 \times 4 \cdot 6$

Mark the sets of vectors which are linearly independent in $\mathbb{R}^3$

☒ a. $v_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ $\times -1$

☒ b. $v_1 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, v_4 = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$

☒ c. $v_1 = \begin{pmatrix} 3 \\ 5 \\ -6 \end{pmatrix}, v_2 = \begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$

☒ d. $v_1 = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$

Mark the vectors which can be obtained as a linear combination of

$$v_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \quad \leftarrow \text{Independent}$$

☒ a. $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} \leftarrow \lambda_1 \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda_3 \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \end{pmatrix}$

☒ b. $\begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$

☒ c. $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

☒ d. $\begin{pmatrix} -7 \\ 1 \\ 4 \end{pmatrix}$

☒ e. $\begin{pmatrix} 0 \\ 9 \\ 0 \end{pmatrix} \leftarrow$

これは a を求めるための!

$$\begin{aligned}2x + y + 3z &= \\ -x + 2y &= \\ y &= \end{aligned}$$

a) $y = 5$
 $-x + 2y = 2 \Rightarrow -x + 10 = 2 \Rightarrow x = 8$
 $2x + y + 3z = 0 \Rightarrow 16 + 5 + 3z = 0 \Rightarrow 3z = -21 \Rightarrow z = -7$

b) ... 全て計算可能だから、a, b, c, d 全部が答え!!

If

$$A = \begin{pmatrix} 4 & 1 & -2 \\ 3 & -2 & 1 \\ -3 & 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} -5 & -2 \\ -2 & 4 \end{pmatrix}$$

then $\det(A) =$ 20 and $\det(B) =$ -24

$$\det(B) = (-20) - (4) = -20 - 4 = -24$$

$$\det(A) = \begin{vmatrix} 4 & 1 & -2 \\ 3 & -2 & 1 \\ -3 & 0 & -1 \end{vmatrix} = 8 - 3 + 0 + 12 - 0 + 3 = 20$$

Mark the sets of vectors which span $\mathbb{R}^2$.

☒ a. $v_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

☒ b. $v_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

☒ c. $v_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, v_2 = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$ $\times -2$

☒ d. $v_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\times 4$ $\times \frac{1}{2}$

$v_1 = \frac{1}{2} v_2 + 4 v_3 \leftarrow v_2 \text{ and } v_3 \text{ are linearly independent } \mathbb{R}^2$

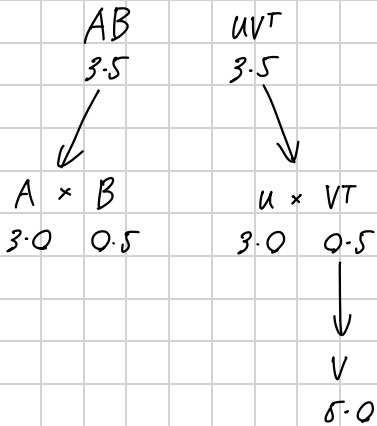
Using some matrices A and B , and some column vectors u and v we obtained, that

$$AB + uv^T = \begin{pmatrix} -2 & -4 & 3 & -4 & -4 \\ -2 & 4 & -4 & -5 & 5 \\ -5 & -5 & -2 & -2 & 5 \end{pmatrix}.$$

Mark the true statements.

Select one or more:

- ☒ a. The product Bv is computable.
 0.5×5.0
- ☒ b. The matrix A has 3 rows.
- ☒ c. The vector u has 3 coordinates.
- ☒ d. The vector v has 3 coordinates.
 5
- ☒ e. The matrix A has as many rows as many columns B has.
 $3 \quad 5$



Give the value of α , if we know that x and y are linearly dependent.

$$x = \begin{pmatrix} 6 \\ 9 \end{pmatrix}, \quad y = \begin{pmatrix} -2 \\ \alpha \end{pmatrix}$$

$9 \div -3 = -3$

Answer:



Mark the sets of vectors which are linearly independent in $\mathbb{R}^3$

☒ a. $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ -2 \\ 0 \end{pmatrix}$, $\times -2$

☒ b. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$,

☒ c. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$,

☒ d. $v_1 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$,

Using some matrices A and B , and some column vectors u and v we obtained, that

$$AB + uv^T = \begin{pmatrix} -3 & 3 & -1 & -3 \\ -2 & 4 & -1 & 1 \\ -3 & 4 & -1 & 5 \\ -1 & 4 & -3 & -1 \\ 3 & 1 & -4 & 3 \\ -1 & -4 & -4 & 2 \end{pmatrix}.$$

Mark the true statements.

Select one or more:

☒ a. The product Bv is computable.

☒ b. The matrix B has 4 columns.

☒ c. The matrix B has 6 rows.

☒ d. The vector v has 4 coordinates.

☒ e. The vectors u and v have the same size.

6.4

AB
6.4
 $\swarrow$
 $A \times B$
6.0 0.4

uv^T
6.4
 $\swarrow$
 $u \times v^T$
6.0 0.4
 $\downarrow$
 v
4.0

Give the value of α , if we know that x and y are linearly dependent.

$$x = \begin{pmatrix} -6 \\ 12 \end{pmatrix}, y = \begin{pmatrix} \alpha \\ 4 \end{pmatrix}$$

$\xrightarrow{\div 3}$

$$-6 \div 3 = -2$$

Answer:

Mark the sets of vectors which span $\mathbb{R}^2$

☒ a. $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, v_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$,

☒ b. $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ -5 \end{pmatrix}, v_3 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}, v_3 = 3v_1 + 0v_2 \rightarrow \mathbb{R}^2$

☒ c. $v_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, $\times -1$

☒ d. $v_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$,

Consider the vectors and a matrix below:

$$x^T = (-10, 1, -10, 3)^T$$

$$y^T = (6, 1, 3, 2)^T$$

$$A = \begin{pmatrix} -3 & -3 & -1 & 4 \\ 4 & -1 & 0 & -1 \end{pmatrix}$$

The value of $x^T y$:

If $z = 2x - 6y$, then $z_4 =$

If $z = Ax$, then $z_2 =$

If $C = AA^T$, then $C_{1,1} =$

$$x^T y = (-10 \ 1 \ -10 \ 3) \begin{pmatrix} 6 \\ 1 \\ 3 \\ 2 \end{pmatrix} = (\cdot) = -60 + 1 - 30 + 6 = -83$$

$$2x - 6y = \begin{pmatrix} -20 \\ 2 \\ -20 \\ 6 \end{pmatrix} + \begin{pmatrix} -6 \\ -6 \\ -18 \\ 12 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{pmatrix} = 6 - 12 = -6$$

$$Ax = \begin{pmatrix} -3 & -3 & -1 & 4 \\ 4 & -1 & 0 & -1 \end{pmatrix} \begin{pmatrix} -10 \\ 1 \\ -10 \\ 3 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} = -40 - 1 + 0 - 3 = -44$$

$$AA^T = \begin{pmatrix} -3 & -3 & -1 & 4 \\ -3 & -3 & -1 & 4 \end{pmatrix} \begin{pmatrix} -3 \\ -3 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 & \cdot \\ \cdot & \cdot \end{pmatrix} = 9 + 9 + 1 + 16 = 35$$

Mark the vectors which can be obtained as a linear combination of

$$v_1 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}, v_3 = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

☒ a. $\begin{pmatrix} -7 \\ 0 \\ 0 \end{pmatrix}$ $\lambda_1 \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} + \lambda_3 \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \cdot \\ \cdot \\ \cdot \end{pmatrix}$ $\leftarrow -7 !!$

☒ b. $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$ same $2x + 3y - z = ?$ $2x + 3y - z = ?$

☒ c. $\begin{pmatrix} -7 \\ 0 \\ -7 \end{pmatrix}$

☒ d. $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ not same

☒ e. $\begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}$ $\leftarrow \times$

Mark the computable expressions.

$$A = \begin{pmatrix} -5 & 4 & -4 & 3 \\ 4 & 2 & 2 & 3 \\ -2 & -5 & 2 & -1 \\ 1 & 1 & 4 & 4 \\ -2 & -4 & -4 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 & 2 & -3 \\ 4 & 5 & -5 & -1 \\ -4 & 1 & 2 & -3 \\ 4 & 1 & -4 & 5 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ -1 \\ -5 \\ 3 \end{pmatrix}, \quad v = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 4 \\ -1 \end{pmatrix}.$$

~~5-4~~ ~~4-4~~ ~~4-1~~ ~~5-1~~

Select one or more:

- ☒ a. AB^T ~~5-4~~ × ~~4-4~~
- ☐ b. $u^T v$ ~~1-4~~ × ~~5-1~~
- ☐ c. Bv ~~4-4~~ × ~~5-1~~
- ☒ d. $v^T A$ ~~1-5~~ × ~~5-4~~
- ☒ e. vu^T ~~5-1~~ × ~~1-4~~

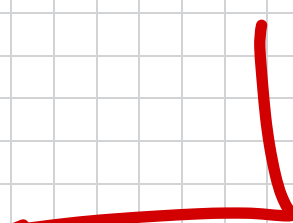
If

$$A = \begin{pmatrix} -4 & -3 & 1 \\ 2 & 1 & -2 \\ -2 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 2 \\ 5 & 2 \end{pmatrix},$$

then $\det(A) =$ and $\det(B) =$

$$\det(B) = (10) - (10) = 0$$

$$\det(A) = \begin{vmatrix} -4 & -3 & 1 \\ 2 & 1 & -2 \\ -2 & 0 & 1 \end{vmatrix} = -4(-2+0) + 2(-0+6) = -8$$



Mark the computable expressions.

$$A = \begin{pmatrix} -3 & -3 & -2 & -2 & -1 \\ -1 & 5 & 1 & -3 & 3 \\ -2 & -2 & -2 & -2 & 4 \\ -5 & 2 & 5 & -2 & -2 \\ 5 & 5 & -4 & 3 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -3 & 4 & 1 & -4 & 3 \\ -3 & -1 & 3 & -4 & 5 \\ -4 & 1 & 2 & 1 & -1 \\ 2 & 2 & -2 & -3 & 3 \\ -4 & 4 & -5 & 4 & -4 \\ -2 & -3 & -3 & 4 & 4 \end{pmatrix}, \quad u = \begin{pmatrix} -4 \\ -5 \\ -4 \\ 5 \\ -2 \end{pmatrix}, \quad v = \begin{pmatrix} -5 \\ 5 \\ -2 \\ 3 \\ -1 \end{pmatrix}.$$

$5 \cdot 5$ $6 \cdot 5$ $5 \cdot 1$ $6 \cdot 1$

Select one or more:

- ☒ a. $v^T u$ $1 \cdot 6 \times 5 \cdot 1$
☒ b. AB^T $5 \cdot 5 \times 5 \cdot 6$
☒ c. BA^T $6 \cdot 5 \times 5 \cdot 5$
☒ d. $v^T v$ $1 \cdot 6 \times 6 \cdot 1$
☒ e. $B^T A$ $5 \cdot 6 \times 5 \cdot 5$

Mark the sets of vectors which are linearly independent in $\mathbb{R}^3$.

☒ a. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$

☒ b. $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix},$

☒ c. $v_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ -4 \\ 0 \end{pmatrix},$

☒ d. $v_1 = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix},$

Consider the vectors and a matrix below:

$$\begin{aligned} x^T &= (3, -7, 3)^T \\ y^T &= (-10, 5, -9)^T \\ A &= \begin{pmatrix} -2 & -3 & 5 \\ -3 & -5 & 5 \end{pmatrix} \end{aligned}$$

The value of $x^T y$:

If $z = 4x - 5y$, then $z_2 =$

If $z = Ax$, then $z_1 =$

If $C = AA^T$, then $C_{1,2} =$

$$x^T y = (3 \ -7 \ 3) \begin{pmatrix} -10 \\ 5 \\ -9 \end{pmatrix} = (\cdot) = -30 - 35 + 27 = -92$$

$$4x - 5y = \begin{pmatrix} -28 \\ -25 \end{pmatrix} + \begin{pmatrix} -25 \\ -25 \end{pmatrix} = \begin{pmatrix} \cdot \\ z_2 \\ \cdot \end{pmatrix} = -28 - 25 = -53$$

$$Ax = \begin{pmatrix} -2 & -3 & 5 \\ -3 & -5 & 5 \end{pmatrix} \begin{pmatrix} 3 \\ -7 \\ 3 \end{pmatrix} = \begin{pmatrix} z_1 \\ \cdot \end{pmatrix} = -6 + 21 + 15 = 30$$

$$AA^T = \begin{pmatrix} -2 & -3 & 5 \\ -3 & -5 & 5 \end{pmatrix} \begin{pmatrix} -3 \\ -5 \\ 5 \end{pmatrix} = \begin{pmatrix} \cdot & C_{1,2} \\ \cdot & \cdot \end{pmatrix} = 6 + 15 + 25 = 46$$

If

$$A = \begin{pmatrix} -4 & -2 & -2 \\ -3 & 4 & 4 \\ -3 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & -4 \\ -3 & 4 \end{pmatrix},$$

then $\det(A) =$ and $\det(B) =$

$$\det(B) = (-8) - (12) = -20$$

$$\det(A) = \begin{vmatrix} -4 & -2 & -2 \\ -3 & 4 & -3 \\ -3 & 0 & 4 \end{vmatrix} = -64 + 24 + 0 - 24 - 0 - 24 = -88$$

Give the value of α , if we know that x and y are linearly dependent.

$$x = \begin{pmatrix} -6 \\ 12 \end{pmatrix}, \quad y = \begin{pmatrix} \alpha \\ 6 \end{pmatrix}$$

$\div 2$
 $-6 \div 2 = -3$

Answer:

Mark the sets of vectors which span $\mathbb{R}^2$.

☒ a. $v_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad v_3 = v_1 - v_2$

☒ b. $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, v_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

☒ c. $v_1 = \begin{pmatrix} 1 \\ -3 \end{pmatrix}, v_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = -v_1 + 0v_3$

☒ d. $v_1 = \begin{pmatrix} 4 \\ -1 \end{pmatrix},$

Using some matrices A and B , and some column vectors u and v we obtained, that

$$AB + uv^T = \begin{pmatrix} 5 & 4 & 4 & 2 \\ -5 & -2 & -4 & -1 \\ 5 & -1 & -2 & 5 \\ -5 & -4 & 2 & -5 \\ -4 & 4 & 2 & -4 \\ -4 & -1 & 1 & 4 \end{pmatrix}.$$

$6 \cdot 4$

Mark the true statements.

Select one or more:

- ☒ a. Matrices A and B have the same number of columns.
☒ b. The vector v has 4 coordinates.
☒ c. The product Bv is computable.
☒ d. The matrix B has 4 columns.
☒ e. The product Au is computable.

$$\begin{array}{cc} AB & uv^T \\ 6 \cdot 4 & 6 \cdot 4 \\ \downarrow & \downarrow \\ A \times B & u \times v^T \\ 6 \cdot 0 \quad 0 \cdot 4 & 6 \cdot 0 \quad 0 \cdot 4 \\ & \downarrow \\ & v \\ & 4 \cdot 0 \end{array}$$

Mark the vectors which can be obtained as a linear combination of

$v_1 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$

☒ a. $\begin{pmatrix} -7 \\ 1 \\ -7 \end{pmatrix}$ *same*

☒ b. $\begin{pmatrix} 0 \\ 9 \\ 0 \end{pmatrix}$ *same*

☐ c. $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

☒ d. $\begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix}$ *same*

☐ e. $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$

$$\lambda_1 \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix}$$

$$\begin{array}{rcl} 2\lambda_1 - \lambda_3 & = & ? \\ 2\lambda_2 & = & ? \\ \cancel{2\lambda_1 - \lambda_3} & = & ? \end{array} \quad \begin{array}{l} \swarrow \\ \searrow \end{array} \text{ same}$$

